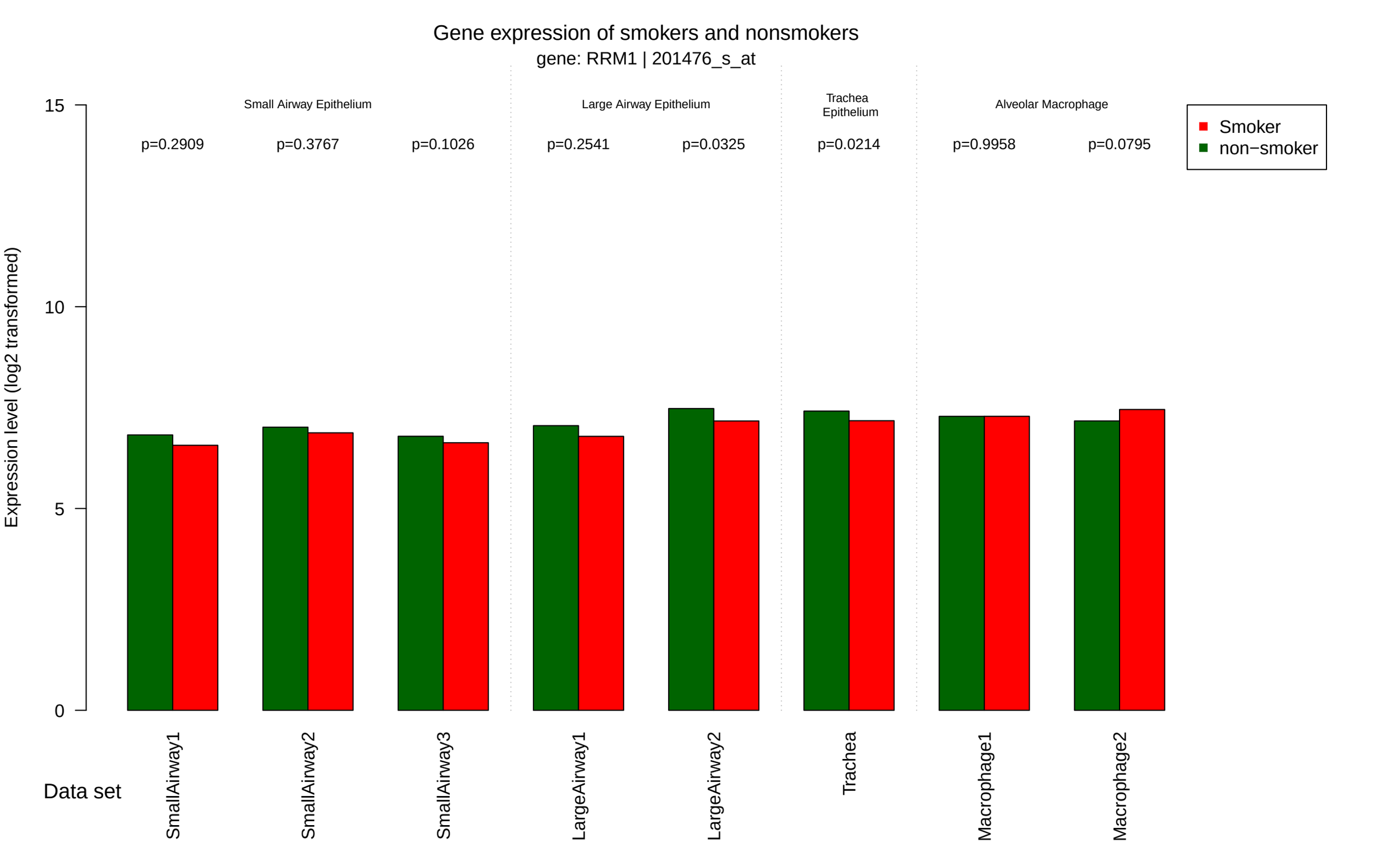
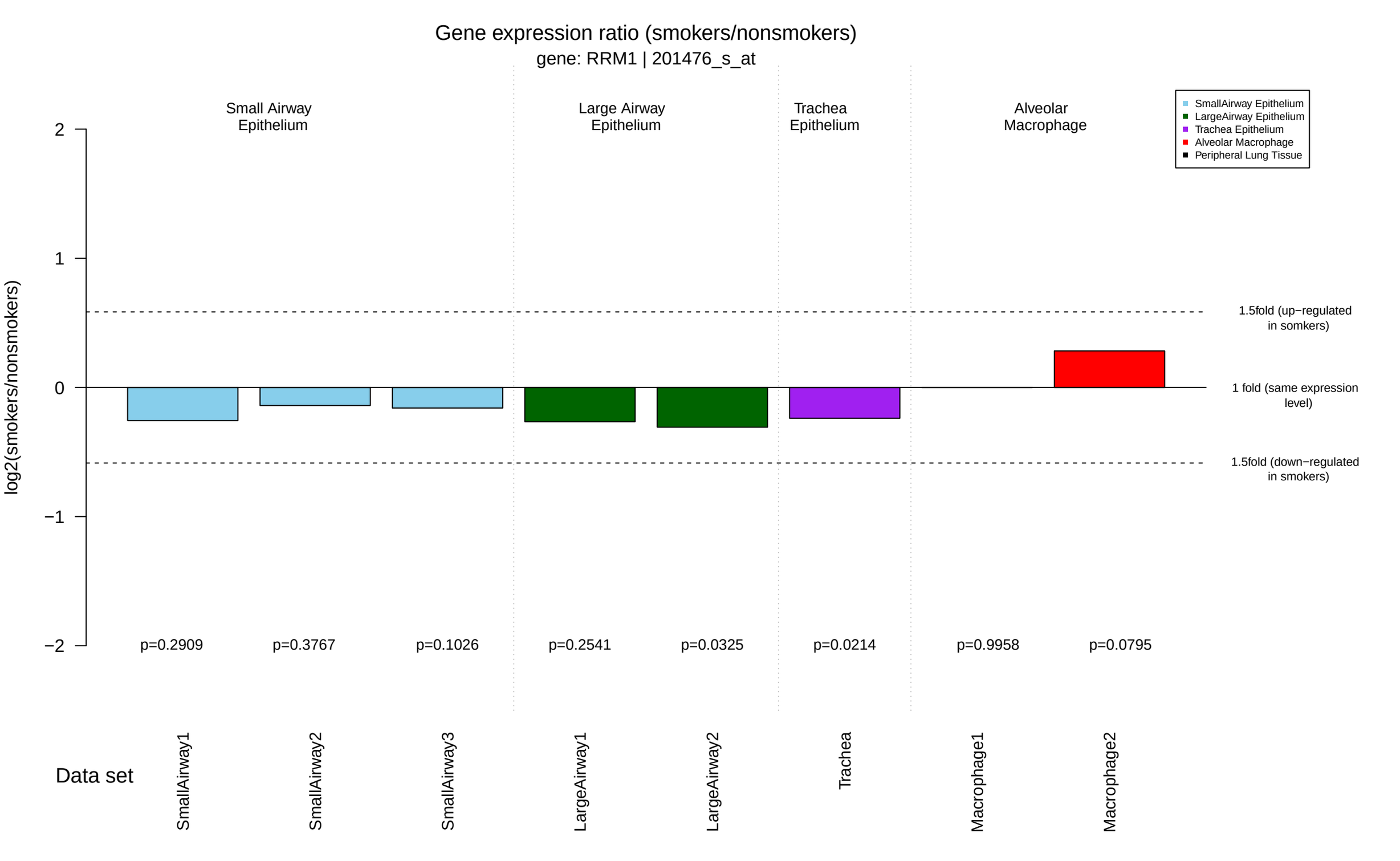
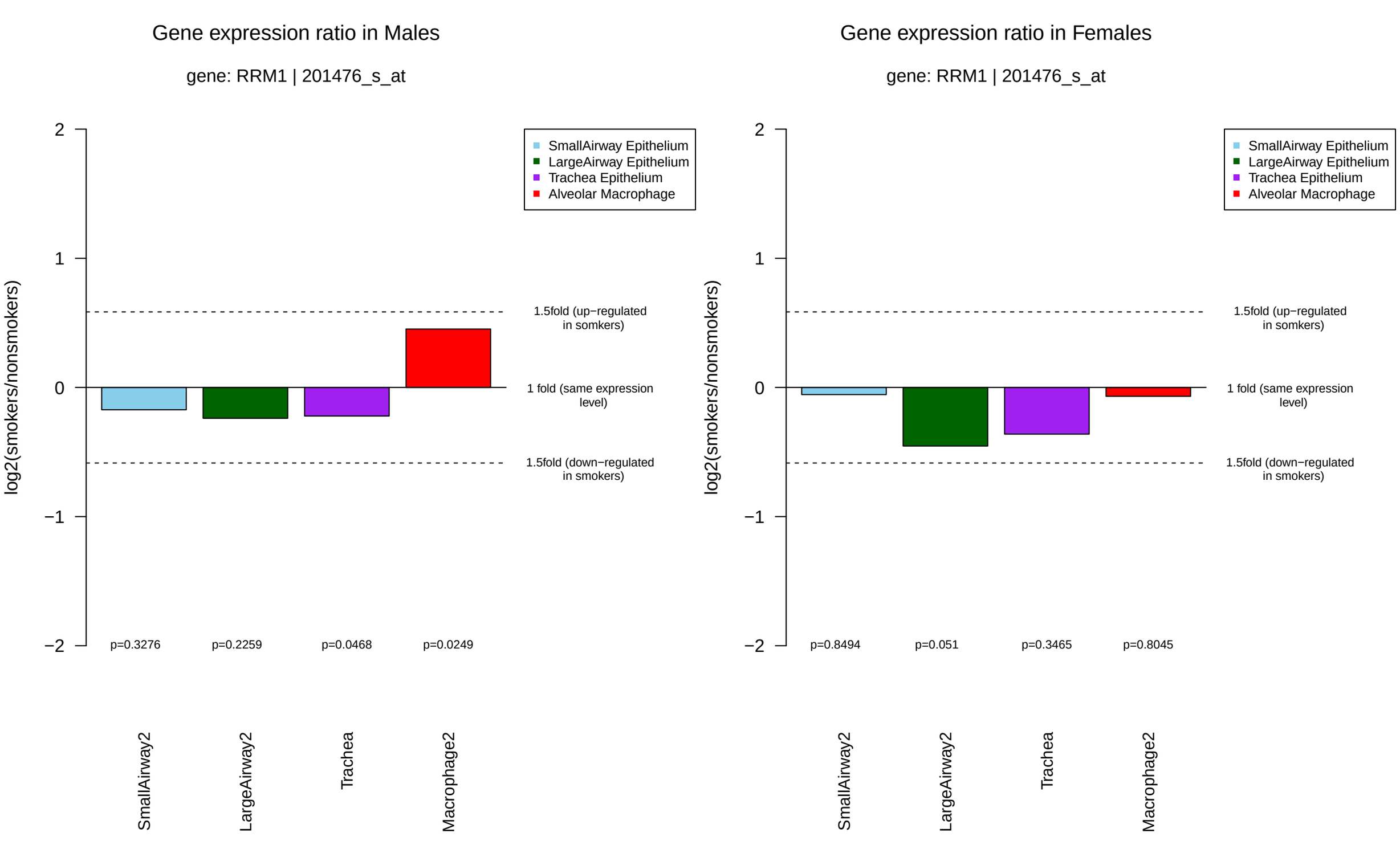
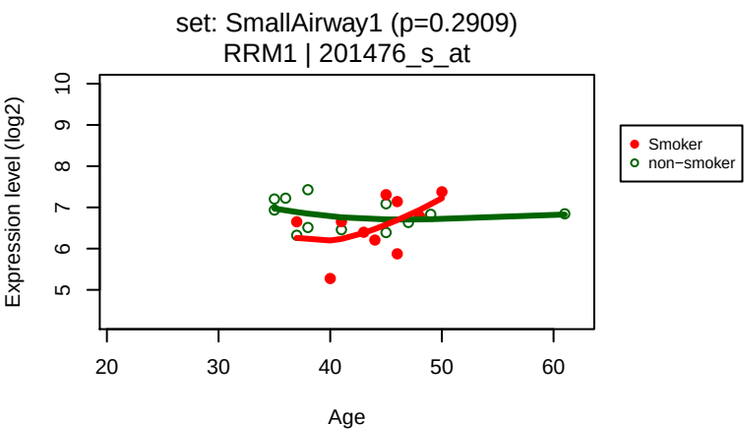
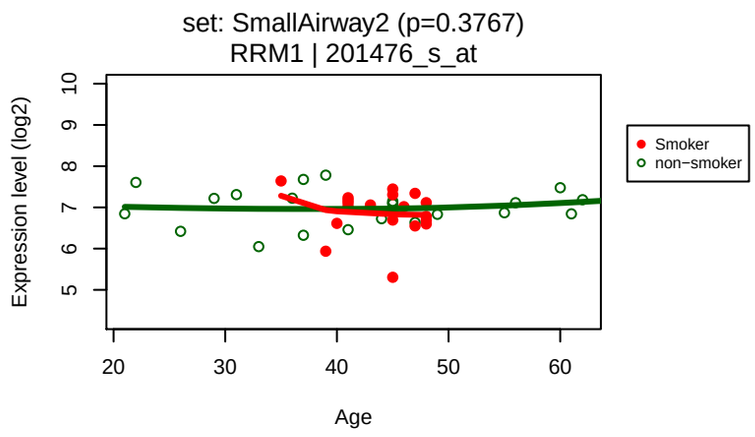
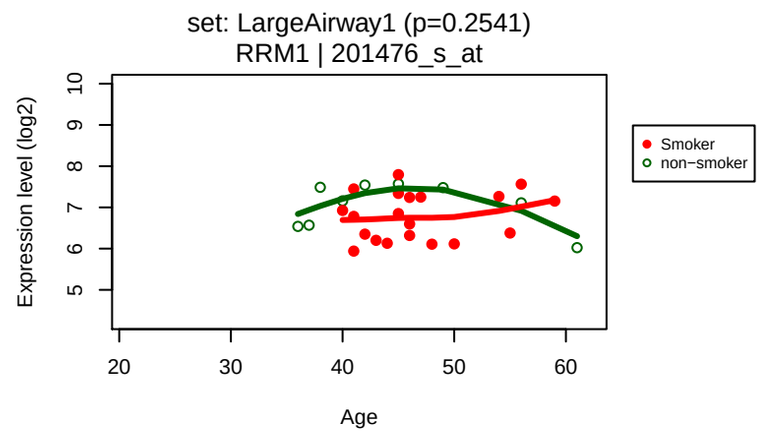

Gene expression of smokers and nonsmokers
gene: RRM1 | 201476_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2909
p=0.3767
p=0.1026
p=0.2541
p=0.0325
p=0.0214
p=0.9958
p=0.0795
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

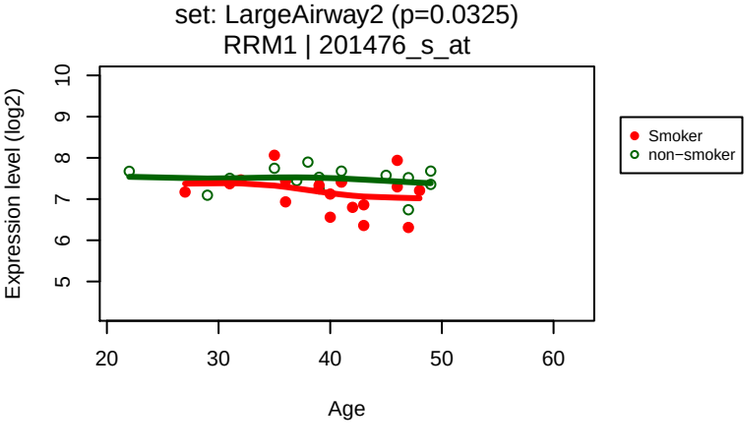

Gene expression ratio (smokers/nonsmokers)
gene: RRM1 | 201476_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2909
p=0.3767
p=0.1026
p=0.2541
p=0.0325
p=0.0214
p=0.9958
p=0.0795
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

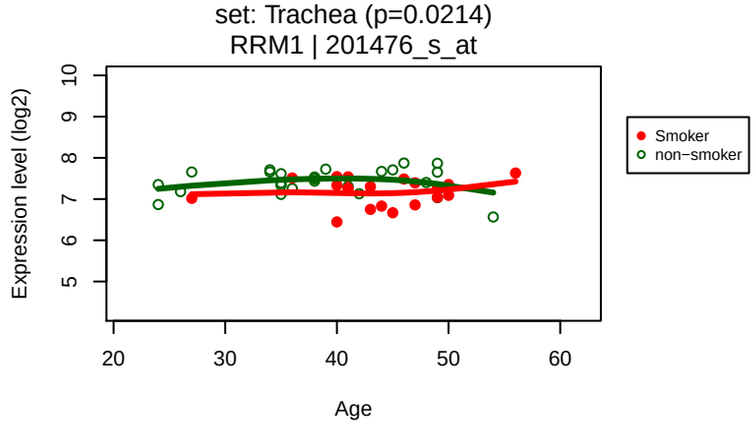

Gene expression ratio in Males
gene: RRM1 | 201476_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3276
p=0.2259
p=0.0468
p=0.0249
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RRM1 | 201476_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8494
p=0.051
p=0.3465
p=0.8045
SmallAirway2
LargeAirway2
Trachea
Macrophage2

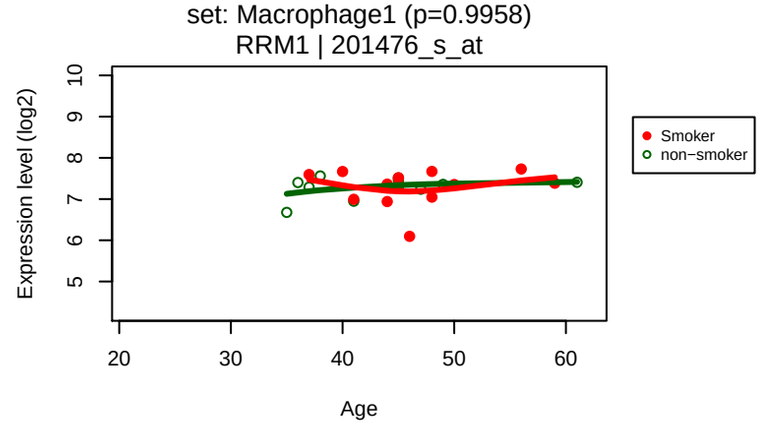

set: SmallAirway1 (p=0.2909)
RRM1 | 201476_s_at
Expression level (log2)
Age
Smoker
non−smoker

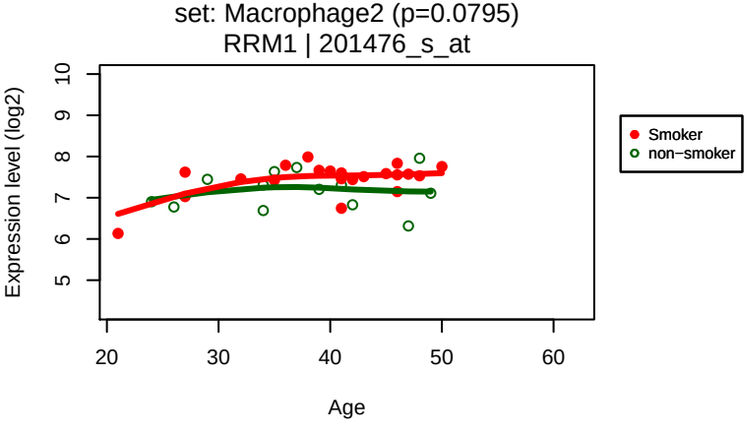

set: SmallAirway2 (p=0.3767)
RRM1 | 201476_s_at
Expression level (log2)
Age
Smoker
non−smoker

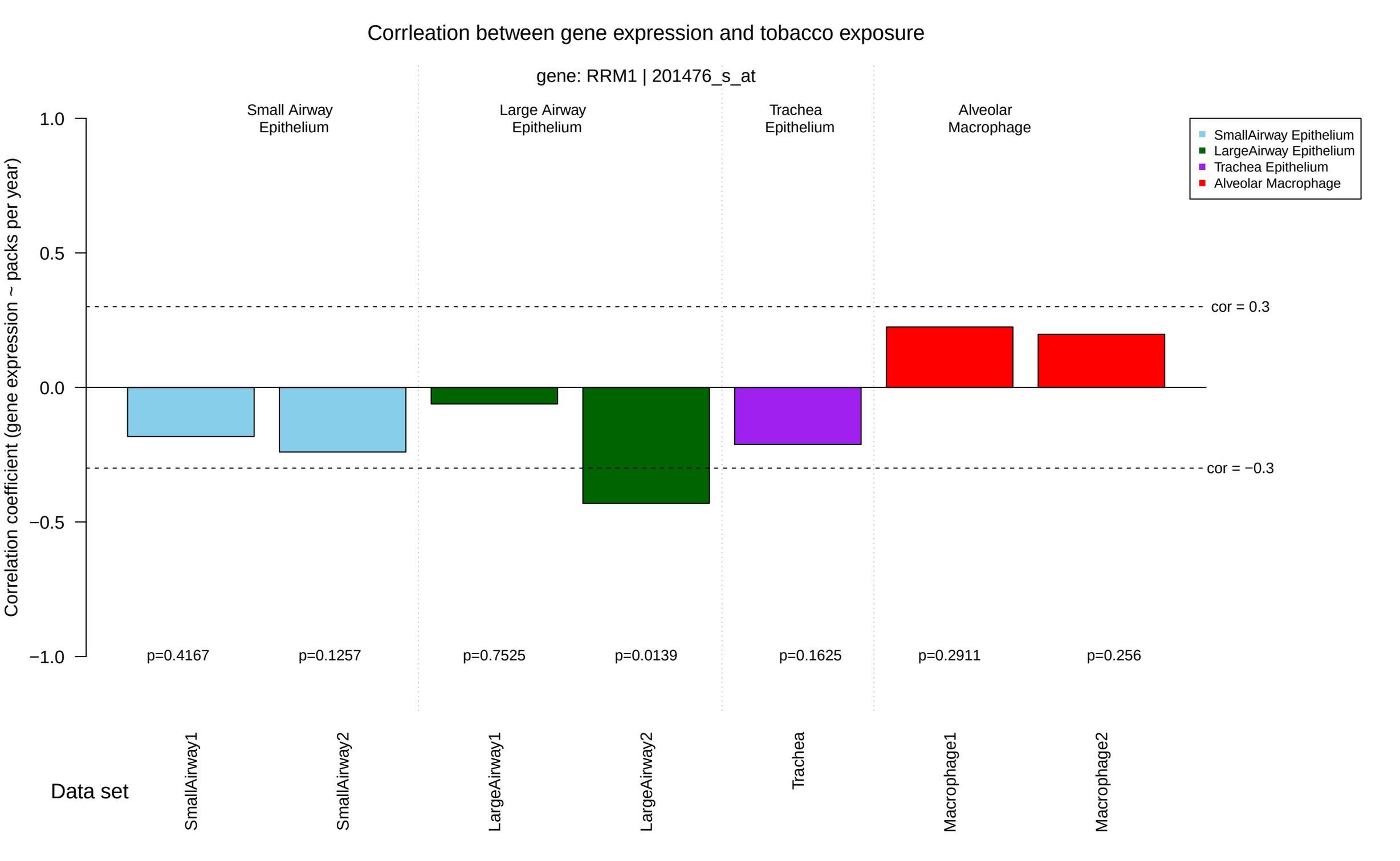

set: LargeAirway1 (p=0.2541)
RRM1 | 201476_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0325)
RRM1 | 201476_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0214)
RRM1 | 201476_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9958)
RRM1 | 201476_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0795)
RRM1 | 201476_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RRM1 | 201476_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4167
p=0.1257
p=0.7525
p=0.0139
p=0.1625
p=0.2911
p=0.256
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2